

# Shock Problem and PINN Framework

## (a) Problem setup

Consider the 1D inviscid Burgers Riemann problem.

**Governing equation**

$$u_t + uu_x = 0, \quad x \in [-1, 1], \quad t \in (0, 1].$$

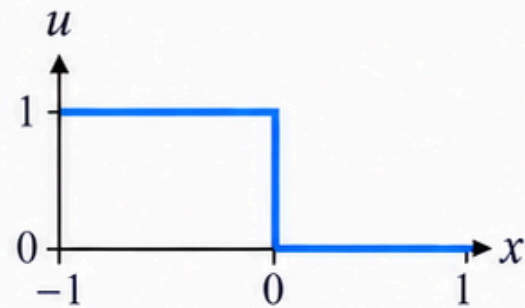
**Initial condition**

$$u(x, 0) = \begin{cases} 1, & x < 0, \\ 0, & x > 0. \end{cases}$$

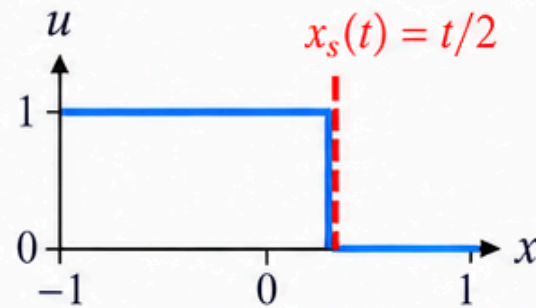
**Boundary condition**

$$u(-1, t) = 1, \quad u(1, t) = 0, \quad t \in (0, 1].$$

Initial condition



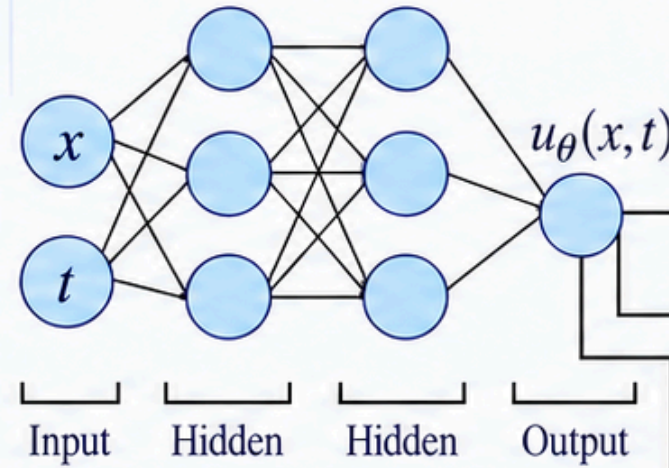
Entropy shock,  $t = 0.5$



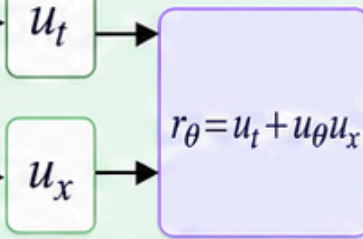
Shock speed:  $(u_L + u_R)/2 = 0.5$ .

## (b) PINN framework

**Coordinate network (schematic)**



**Automatic differentiation**



**Loss computation**

**PDE loss**

$$\mathcal{L}_{\text{PDE}} = \frac{1}{N_f} \sum_{j=1}^{N_f} |r_{\theta}(x_f^j, t_f^j)|^2$$

**Initial condition loss**

$$\mathcal{L}_{\text{IC}} = \frac{1}{N_i} \sum_{j=1}^{N_i} |u_{\theta}(x_i^j, 0) - u_0(x_i^j)|^2$$

**Boundary condition loss**

$$\mathcal{L}_{\text{BC}} = \frac{1}{N_b} \sum_{j=1}^{N_b} |u_{\theta}(x_b^j, t_b^j) - g(x_b^j, t_b^j)|^2$$

$$\mathcal{L} = \mathcal{L}_{\text{PDE}} + \mathcal{L}_{\text{IC}} + \mathcal{L}_{\text{BC}}$$

Update  $\theta$

**Training: backpropagation / optimizer**

$$\theta \leftarrow \theta - \eta \nabla_{\theta} \mathcal{L}$$